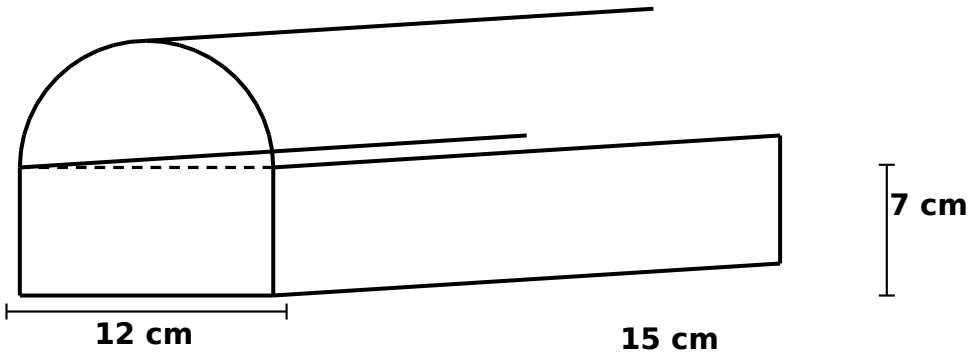
Surface Area of 3D Shapes

Name: _____ Date: _____

Read one question at a time. Use the diagram and the labelled measurements. You may use a ruler, formula sheet or extra scrap paper. Show your mathematical thinking using numbers, equations, labels or diagrams.

Q5 Composite prism

The cross-section is made by joining a semicircle and a rectangle. Calculate the total surface area of the prism. Give your answer to 3 significant figures.



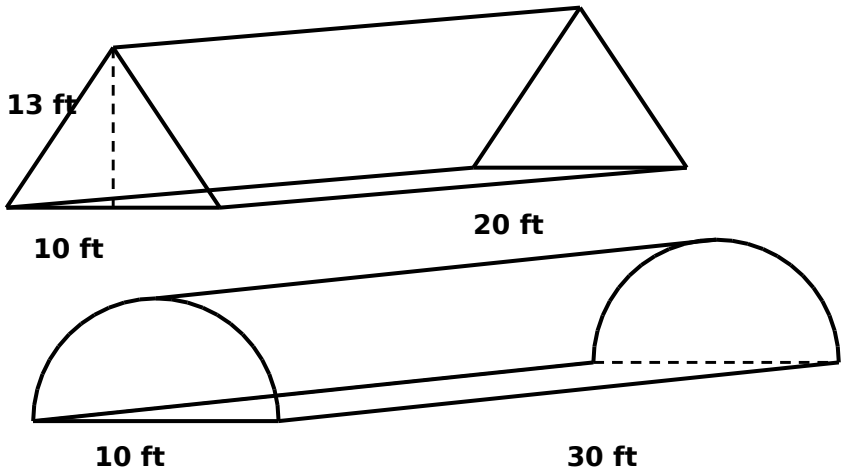
Helpful reminder: The semicircle's diameter is 12 cm. The prism is 15 cm long. The rectangular part is 7 cm high.

My working

Surface Area of 3D Shapes

Q6 Compare two solids

A triangular prism and a semi-cylinder are shown. Which solid has the larger total surface area? Calculate both surface areas, then compare them.

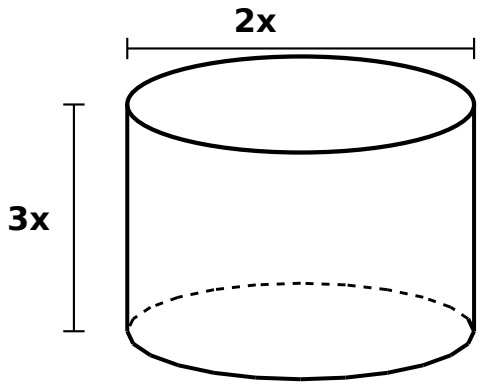


Measurements: Triangular prism: equal sloping sides 13 ft, base 10 ft, length 20 ft. Semi-cylinder: diameter 10 ft, length 30 ft.

My working

Q7 Solve for x

The cylinder has a total surface area of 392π cm². The diameter is $2x$ and the height is $3x$. Find x.

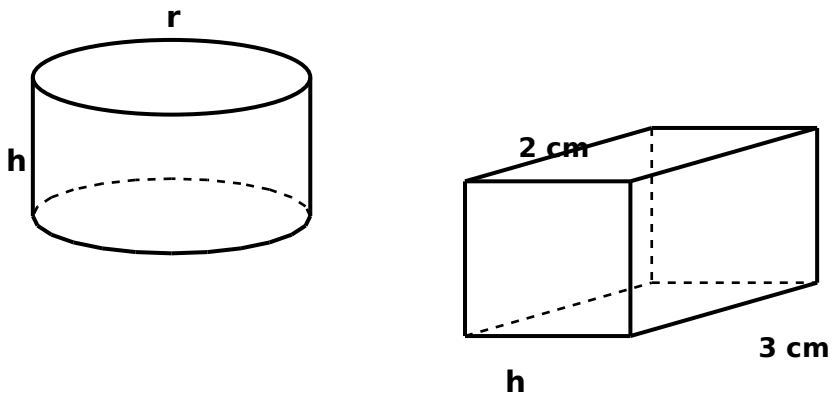


My working

Surface Area of 3D Shapes

Q8 Equal surface areas

The cylinder and cuboid have the same total surface area. Express h in terms of r.



Measurements: Cylinder radius = r and height = h. Cuboid dimensions = h by 2 cm by 3 cm.

My working

Before you finish

☐ I used the correct measurements and units.
☐ I included every outside face.
☐ I checked that my final answer answers the question.